Software Engineering

Assignment-7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

QUESTION 1:

Which of the following statements correctly describe the purpose and characteristics of UML models?

a. UML models help manage system complexity by representing every implementation detail of the system.
b. Different UML models may emphasize different aspects of a system
c. UML primarily represents low-level program execution details only
d. UML is primarily a graphical modelling technique
e. UML modelling captures only the important aspects of a system while ignoring unnecessary details
f. UML models are generally more compact and easier to understand than textual system descriptions

Correct Answer: b., d., e., and f.

Detailed Solution:

UML uses graphical models to represent important system aspects at different levels of abstraction. Different diagrams emphasize different views and are generally easier to understand than lengthy textual descriptions.

QUESTION 2:

Consider the statement: “A savings account is a type of bank account.” Assuming BankAccount and SavingsAccount are two classes, which of the following relationships best describes this situation?

a. Aggregation
b. Association
c. Generalization-specialization
d. Composition
e. Overloading
f. Dependency

Correct Answer: c. Generalization-specialization

Detailed Solution:

A saving account is a type of bank account is an example of Generalization-specialization.

QUESTION 3:

Which of the following statements correctly describe multiple inheritance in object-oriented programming?

a. A base class has multiple derived classes
b. A class is derived from more than one parent class
c. Multiple inheritance represents a situation where two classes recursively inherit from each other
d. A derived class can combine functionalities inherited from multiple parent classes
e. A class derived through several levels of inheritance is an example of multiple inheritance

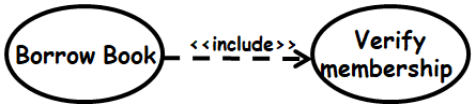
Correct Answer: b. A class is derived from more than one parent class
d. A derived class can combine functionalities inherited from multiple parent classes

Detailed Solution:

Multiple inheritance means a class inherits from more than one parent class. It allows a derived class to combine features from multiple base classes.

QUESTION 4:

Which of the following statements are correct regarding the following diagram?



a. The Borrow Book use case always performs Verify Member
b. Verify Member represents common functionality reusable by other use cases
c. The relationship shown is an example of use-case generalization
d. The included use case helps reduce duplication in system behavior
e. Verify Member is executed only under exceptional conditions
f. The diagram represents an «include» relationship between use cases

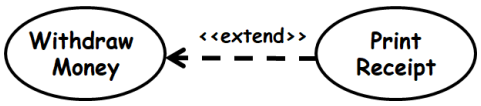
Correct Answer: a. The Borrow Book use case always performs verify member, **b.** Verify member represents common functionality reusable by other use cases, **d.** The included use case helps reduce duplication in system behavior, **f.** The diagram represents an <<include>> relationship between use cases

Detailed Solution:

An «include» relationship means the included use case is always executed and represents reusable common functionality. It helps reduce duplication among use cases.

QUESTION 5:

Consider the following UML use-case diagram:



Which of the following statements are correct regarding the above diagram?

a. Withdraw Cash provides optional behavior to Print Receipt
b. The base use case Withdraw Cash can execute without Print Receipt
c. The «extend» relationship is used to model conditional or optional functionality
d. Withdraw Cash extends Print Receipt
e. Print Receipt is always executed whenever Withdraw Cash occurs
f. The extending use case is invoked only under specific conditions

Correct Answer: b. The base use case Withdraw Cash can execute without Print Receipt, **c**. The «extend» relationship is used to model conditional or optional functionality, **f.** The extending use case is invoked only under specific conditions

Detailed Solution:

An «extend» relationship models optional/conditional behavior. The base use case can execute independently, while the extending use case runs only under specific conditions.

QUESTION 6:

In an online shopping system, before a customer can place an order, the system must verify whether the payment details are valid. Which of the following UML use-case relationships is most appropriate to factor the Place Order use case into Validate Payment functionality?

a. Aggregation
b. Include
c. Extend
d. Inheritance
e. Dependency
f. Composition

Correct Answer: b. Include

Detailed Solution:

Validate Payment is mandatory before Place Order can complete. The «include» relationship is used for mandatory reusable functionality.

QUESTION 7:

Which of the following are NOT characteristics of a good textual description accompanying a UML use-case diagram?

a. The textual description should be understandable from the actor's perspective
b. The description should be written entirely using programming language syntax
c. The description should focus on interactions between the actor and the system
d. The use-case description should primarily describe message exchanges among classes
e. The description should conclude after the system responds to the actor's request
f. The textual description should rely mainly on internal implementation details familiar only to developers

Correct Answer: b. The description should be written entirely using programming language syntax

d. The use-case description should primarily describe message exchanges among classes

f. The textual description should rely mainly on internal implementation details familiar only to developers

Detailed Solution:

Use-case descriptions should be written in simple user-oriented language focusing on actor-system interaction, not programming syntax, class-level details, or implementation internals.

QUESTION 8:

Which of the following statements are FALSE regarding UML models of a software system?

a. UML models help manage system complexity through abstraction
b. Every UML model represents the design view of a system only
c. UML design diagrams are models of the system
d. UML models are useful only after the software has been fully implemented.
e. Dynamic UML models represent behavioral aspects of the system
f. Static UML models represent structural aspects of the system

Correct Answer: b. Every UML model represents the design view of a system only, **d.** UML models are useful only after the software has been fully implemented

Detailed Solution:

UML models are not limited to the design view only; they can represent analysis, design, and other perspectives. UML models are useful throughout development, not only after implementation.

QUESTION 9:

For a class hierarchy to support runtime polymorphism in an object-oriented system, which of the following relationships or features are generally required?

a. Inheritance
b. Aggregation
c. Method overriding
d. Generalization
e. Dependency

Correct Answer: a. Inheritance
c. Method overriding
d. Generalization

Detailed Solution:

Runtime polymorphism requires inheritance/generalization and method overriding so that derived class methods can be invoked through base class references.

QUESTION 10:

Which of the following are important advantages of using inheritance in object-oriented design?

a. Code reusability
b. Increased duplication of code
c. Support for extensibility
d. Reduced maintainability
e. Hierarchical classification of classes

Correct Answer: a. Code reusability
c. Support for extensibility
e. Hierarchical classification of classes

Detailed Solution:

Inheritance promotes code reuse, supports easy extension of existing classes, and helps organize classes into hierarchical structures.

**********END**********